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L^p estimates for an oscillating Dunkl multiplier

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Abstract

In this paper, we study the L^p boundedness of a class of oscillating multiplier operator for the Dunkl transform, $T_{m_\alpha} = \mathcal{F}_k^{-1}(m_\alpha \mathcal{F}_k(f))$ with $m(\xi) = |\xi|^{-\alpha} e^{\pm i|\xi|}\phi(\xi)$. We obtain an L^p-bound result for the corresponding maximal functions. As a specific applications, we give an extension of the L^p estimate for the wave equation and of Stein's theorem for the analytic family of maximal spherical means [13].

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1 Introduction and Statement of Results

Let $\alpha > 0$ and $\beta > 0$, the oscillating multiplier $T_{\alpha,\beta}$ is defined via Fourier transform by $\widehat{T_{\alpha,\beta}(f)} = m_{\alpha,\beta}\widehat{f}$, where $m_{\alpha,\beta} = |\xi|^{-\alpha} e^{i|\xi|^\beta}\phi(\xi)$ and ϕ is a C^∞ function on $\mathbb{R}^n$ which vanishes near the origin and is equal to 1 for all sufficiently large ξ. The study of their L^p properties going back to the works of I. Hirschman [7] in the case $n = 1$ and S. Wainger [15] in higher dimensions. Later on, they have been extensively studied again by many authors in several different contexts, see [6, 8, 9, 11]. This paper is devoted to the study of L^p boundedness of oscillating multiplier in the context of Dunkl analysis. We will focus on the case $\beta = 1$, because of the close connection to the wave equation associated with the Dunkl Laplacian Δ_k, and to the spherical maximal function. The latter is already studied by L. Deleaval [4]. In order to describe more precisely the results studied in this paper, we shall start by giving a brief summary of the Dunkl analysis.

1.1 Background

Dunkl theory generalizes classical Fourier analysis on $\mathbb{R}^n$. It started twenty years ago with Dunkl's seminal work [5] and was further developed by several mathematicians. We refer for more details to the articles [5, 3, 10] and the references cited therein.

Let $G \subset \mathrm{O}(\mathbb{R}^n)$ be a finite reflection group associated to a reduced root system R and $k : R \to [0, +\infty)$ a G–invariant function (called multiplicity function). Let R^+ be a positive root subsystem. The Dunkl operators D^k_ξ on $\mathbb{R}^n$ are the following k–deformations of directional derivatives ∂_ξ by difference operators:

$$D^k_\xi f(x) = \partial_\xi f(x) + \sum_{v \in R^+} k(v)\, \langle v, \xi \rangle\, \frac{f(x) - f(\sigma_v.\, x)}{\langle v,\, x \rangle}\,,$$

where $\sigma_v.\, x = x - \frac{\langle v, x \rangle}{2\,|\alpha|^2}\, v$ denotes the reflection with respect to the hyperplane orthogonal to v. Here $\langle ., . \rangle$ is the usual Euclidean inner product and $|\,.\,|$ its induced norm. The Dunkl operators are antisymmetric with respect to the measure $w_k(x)\, dx$ with density

$$w_k(x) = \prod_{v \in R^+} |\, \langle v, x \rangle\, |^{\,2\,k(v)}\,.$$

The operators ∂_ξ and D^k_ξ are intertwined by a Laplace–type operator

$$V_k f(x) \;= \int_{\mathbb{R}^n} f(y)\, d\mu_x(y)$$

associated to a family of compactly supported probability measures $\{\, \mu_x \,|\, x \in \mathbb{R}^n \,\}$. Specifically, μ_x is supported in the the convex hull $\mathrm{co}(G.x)$.

For every $y \in \mathbb{C}^n$, the simultaneous eigenfunction problem

$$D^k_\xi f = \langle y, \xi \rangle\, f \qquad \forall\, \xi \in \mathbb{R}^n$$

has a unique solution $f(x) = E_k(x, y)$ such that $E_k(0, y) = 1$, called the Dunkl kernel and is given by

$$E_k(x, y) \;=\; V_k(e^{\,\langle ., \, y \rangle})(x) \;=\; \int_{\mathbb{R}^n} e^{\,\langle z, y \rangle}\, d\mu_x(z) \qquad \forall\, x \in \mathbb{R}^n.$$

Furthermore this kernel has a holomorphic extension to $\mathbb{C}^n \times \mathbb{C}^n$ and the following estimate hold: for $x,\ y \in \mathbb{C}^n$,

(ii) $E_k(x, y) = E_k(y, x)$,

(iii) $E_k(\lambda x, y) = E_k(x, \lambda y)$, for $\lambda \in \mathbb{C}$,

(iv) $E_k(g.x, g.y) = E_k(x, y)$, for $g \in G$.

In dimension $n = 1$, these functions can be expressed in terms of Bessel functions. Specifically,

$$E_k(x, y) = \mathcal{J}_{k-\frac{1}{2}}(ixy) + \frac{xy}{2k+1}\, \mathcal{J}_{k+\frac{1}{2}}(ixy)\,,$$

where

$$\mathcal{J}_\nu(z) \;=\; \Gamma(\nu+1) \sum_{n=0}^{+\infty} \frac{(-1)^n}{n!\, \Gamma(\nu + n + 1)} \left(\frac{z}{2}\right)^{2n}$$

are normalized Bessel functions.

The Dunkl transform is defined on $L^1(\mathbb{R}^n, w_k(x)dx)$ by

$$\mathcal{F}_k f(\xi) = c_k \int_{\mathbb{R}^n} f(x)\, E_k(x, -i\,\xi)\, w_k(x)\, dx\,,$$

where

$$c_k \;= \int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2}}\, w(x)\, dx\,.$$

We list some known properties of this transform :

(i) The Dunkl transform is a topological automorphism of the Schwartz space $\mathcal{S}(\mathbb{R}^n)$.

(ii) (*Plancherel Theorem*) The Dunkl transform extends to an isometric automorphism of $L^2(\mathbb{R}^n, w_k(x)dx)$.

(iii) (*Inversion formula*) For every $f \in \mathcal{S}(\mathbb{R}^n)$, and more generally for every $f \in L^1(\mathbb{R}^n, w_k(x)dx)$ such that $\mathcal{F}_k f \in L^1(\mathbb{R}^n, w_k(\xi)d\xi)$, we have

$$f(x) = \mathcal{F}_k^2 f(-x) \qquad \forall\, x \in \mathbb{R}^n.$$

(iv) If f is a radial function in $L^1(\mathbb{R}^n, w_k(\xi)d\xi)$ such that $f(x) = \widetilde{f}(|x|)$, $x \in \mathbb{R}^n$. Then $\mathcal{F}_k(f)$ is also radial and

$$\mathcal{F}_k(f)(x) = d_k \int_0^\infty \widetilde{f}(s)\mathcal{J}_{\gamma_k+n/2-1}(s|x|)s^{2\gamma_k+n}ds; \quad x \in \mathbb{R}^n. \tag{1.1}$$

where $d_k = 2^{-(\gamma_k+n/2-1)}/\Gamma(\gamma_k + n/2)$.

Let $x \in \mathbb{R}^n$, the Dunkl translation operator τ_x is given for $f \in L^2_k(\mathbb{R}^n, w_k(x)dx)$ by

$$\mathcal{F}_k(\tau_x(f))(y) = \mathcal{F}_k f(y)\, E_k(x, iy), \quad y \in \mathbb{R}^n.$$

In the case when $f(x) = \widetilde{f}(|x|)$ is a radial function in $\mathcal{S}(\mathbb{R}^n)$, the Dunkl translation is represented by the following integral

$$\tau_x(f)(y) = \int_{\mathbb{R}^n} \widetilde{f}\left(\sqrt{|y|^2 + |x|^2 + 2\langle y, \eta\rangle}\,\right) d\mu_x(\eta),$$

This formula shows that the Dunkl translation operators can be extended to all radial functions f in $L^p(\mathbb{R}^n, w_k(x)dx)$, $1 \le p \le \infty$ and the following holds

$$\|\tau_x(f)\|_{p,k} \le \|f\|_{p,k}\,. \tag{1.2}$$

where $\|\,.\,\|_{p,k}$ denotes the norm of $L^p(\mathbb{R}^n, w_k(x)dx)$, $1 \le p < \infty$.

We define the Dunkl convolution product for suitable functions f and g by

$$f *_k g(x) = \int_{\mathbb{R}^N} \tau_x(f)(-y)g(y)d\mu_k(y), \quad x \in \mathbb{R}^N.$$

We note that it is commutative and satisfies the following property:

$$\mathcal{F}_k(f *_k g) = \mathcal{F}_k(f)\mathcal{F}_k(g). \tag{1.3}$$

Moreover, the operator $f \to f *_k g$ is bounded on $L^p(\mathbb{R}^n, w_k(x)dx)$ provide g is a bounded radial function in $L^1(\mathbb{R}^n, w_k(x)dx)$. In particular we have the the following Young's inequality:

$$\|f *_k g\|_{p,k} \leq \|g\|_{1,k}\|f\|_{p,k}\ . \tag{1.4}$$

We now come to the subject of this paper.

1.2 Statement of Results

Let ϕ be a C^∞ radial function which vanishes for $|\xi| \leq 1/2$ and is equal to 1 for $|\xi| \geq 1$. Let $\alpha > 0$, we put

$$m_\alpha(\xi) = |\xi|^{-\alpha} e^{\pm i|\xi|}\phi(\xi).$$

The oscillating Dunkl multiplier associated to m_α is defined on $L^2(\mathbb{R}^n, w_k(x)dx)$ by

$$T_{m_\alpha}(f) = \mathcal{F}_k^{-1}(m_\alpha \mathcal{F}_k(f)).$$

We will establish the following.

Theorem 1.1. *T_{m_α} is a bounded operator on $L^p(\mathbb{R}^n, w_k(x)dx)$ for all $1 \leq p \leq \infty$ such that*

$$\alpha > (2\gamma_k + n - 1)\left|\frac{1}{2} - \frac{1}{p}\right|.$$

The proof follows closely the proof given by S. Sjostrand in [11]. This contains L^p estimates of the solution operator for the Cauchy problem associated to Dunkl wave equation

$$\Delta_k u(x,t) = \partial_t^2 u(x,t), \quad u(x,y) = 0, \quad \partial_t u(x,o) = f(x)$$

where $\Delta_k = \sum_{i=1}^n (T_{e_i}^k)^2$, which is referred to as the Dunkl-Laplace operator on $\mathbb{R}^n$. Note that the solution to this problem has been already described in [1] and is given by means of Dunkl's transform,

$$\mathcal{F}_k(u(.,t))(\xi) = \frac{\sin t|\xi|}{|\xi|}\mathcal{F}_k(f)(\xi).$$

As the result, we have

Theorem 1.2. *For fixed t, the linear operator $f \to u(x,t)$ is bounded operator on $L^p(\mathbb{R}^n, w_k(x)dx)$ to itself, for $1 \leq p \leq \infty$ such that*

$$\left|\frac{1}{2} - \frac{1}{p}\right| < \frac{1}{2\gamma_k + n - 1}.$$

Theorem 1.4 is therefore the special case $\alpha = 1$ of theorem 1.2. Indeed, we just write

$$\frac{\sin(|\xi|)}{|\xi|} = \frac{e^{i|\xi|}}{2i|\xi|}\phi(\xi) - \frac{e^{-i|\xi|}}{2i|\xi|}\phi(\xi) + \frac{\sin(|\xi|)}{|\xi|}(1-\phi(\xi)) = a_1(\xi) + a_2(\xi) + a_3(\xi).$$

As a_3 is radial C^∞-function with compact support, the Dunkl multiplier associated to a_3 is the convolution operator with kernel $\mathcal{F}_k^{-1}(a_3)$ which is a radial Schwartz function and then by Young's inequality it is a bounded operator on $L^p(\mathbb{R}^n, w_k(x)dx)$ for all $1 \leq p \leq \infty$.

Remark 1.3. *In the statements of theorems 1.1 and 1.2 nothing can be said about the endpoint of the range of p. Adaptation of classical arguments as in [6, 8, 9] is not available, in the Dunkl setting the theory of Hardy space H^1 and BMO space are not yet much elaborated. Also the L^p theory of Dunkl multiplier is still ambiguous, since the latter is closely related to the generalized translation operators which require more information than its known properties, in particular about their integral representations.*

The next main result is the boundedness of the maximal operator

$$A_\alpha(f)(x) = \sup_{t>0} |A_\alpha(f)(x,t)|,$$

where

$$A_\alpha(f)(x,t) = \int_{\mathbb{R}^n} \frac{e^{\pm it|\xi|}}{|t\xi|^k}\phi(t\xi)\mathcal{F}(f)(\xi)E_k(ix,\xi)w_k(\xi)d\xi.$$

Theorem 1.4. *The maximal operator A_α is bounded on $L^p(\mathbb{R}^n, w_k(x)dx)$, provide*

$$\alpha > (2\gamma_k + n - 1)\left|\frac{1}{p} - \frac{1}{2}\right| + \frac{1}{p}$$

Our arguments inspired by [12] for the study of the boundedness of certain Fourier integral operators. As an application, we provide an extension of the result in [13], obtaining boundedness for the maximal function $\mathcal{M}_\alpha(f)(x) = \sup_{t>o} |\mathcal{M}_\alpha(f)(x,t)|$ where

$$\mathcal{M}_\alpha(f)(x,t) = \int_{\mathbb{R}^n} \mathcal{J}_{\alpha+\gamma_k+n/2-1}(t|\xi|)\mathcal{F}_k(f)(\xi)w_k(\xi)d\xi$$

In particular, $\mathcal{M}_0$ reduces to the analogous of Stein's spherical maximal function,

$$\mathcal{M}_0(f)(x) = \sup_{t>0}\left|\int_{S^{n-1}} \tau_x(f)(ty)d\sigma(y)\right|$$

where S^{n-1} denotes the standard unit sphere in $\mathbb{R}^n$ and $d\sigma$ corresponds to the normalized surface measure. We obtain the following

Theorem 1.5. *The maximal operator $\mathcal{M}_\alpha$ is bounded on $L^p(\mathbb{R}^n, w_k(x)dx)$ under the following conditions*

(a) $\alpha > 1 - 2\gamma_k - n + (2\gamma_k + n)/p, \qquad if \quad 2 \leq p < \infty,$

(b) $\alpha > (2-2\gamma_k-n)/p$, *if* $1 < p \leq 2$.

As a direct consequence of Theorem 1.5 we get the following result.

Corollary 1.6. *Stein's spherical maximal function* $\mathcal{M}_0$ *is bounded on* $L^p(\mathbb{R}^n, w_k(x)dx)$, $n \geq 3$, *provide*

$$p > \frac{2\gamma_k + n}{2\gamma_k + n - 1}\,.$$

2 Details of the Proofs

We begin by choosing the Littlewood-Paley dyadic decomposition of unity that we shall use it all along this section. The existence of such a partition is standard, it is given by a C^∞-function ψ that is supported in $\{t \in \mathbb{R};\ 1/2 \leq |t| \leq 2\}$ and satisfies

$$\sum_{-\infty}^{\infty} \psi(2^{-\nu}t) = 1, \qquad t \neq 0.$$

We can assume further that $0 \leq \psi \leq 1$.

For convenience, we use the same notation C to denote the different constants in the different place.

2.1 Proof of Theorem 1.1

Without loss of generality we will prove the theorem 1.1 by taking ϕ the function

$$\phi(\xi) = \widetilde{\phi}(|\xi|) = \sum_{\nu=0}^{\infty} \psi(2^{-\nu}|\xi|), \tag{2.1}$$

which is clearly a C^∞-function on $\mathbb{R}^n$, with $\phi(\xi) = 0$ for $|\xi| \leq 1/2$ and $\phi(\xi) = 1$ for $|\xi| \geq 1$. Indeed, let ϕ_1 be an arbitrary C^∞-function on $\mathbb{R}^n$, with $\phi_1(\xi) = 0$ for $|\xi| \leq 1/2$ and $\phi_1(\xi) = 1$ for $|\xi| \geq 1$. We write

$$m_\alpha(\xi) = |\xi|^{-\alpha} e^{i|\xi|}(\phi_1(\xi) - \phi(\xi)) + |\xi|^{-\alpha} e^{i|\xi|}\phi(\xi) = m_\alpha^{(1)} + m_\alpha^{(2)}.$$

Since $m_\alpha^{(1)}$ is a C^∞-function with compact support, then the corresponding multiplier is a bounded linear operator on $L^p(\mathbb{R}^n, w_k(x)dx)$, $1 \leq p \leq \infty$.

Assume as a first step that $\alpha \in]\gamma_k + (n-1)/2, \gamma_k + (n+1)/2[$. Here we shall prove that T_{m_α} is a convolution operator with kernel K_α belongs to $L^1(\mathbb{R}^n, w_k(x)dx)$.

We begin, by writing via (2.1)

$$m_\alpha(\xi) = \sum_{\nu=0}^{\infty} m_\alpha^\nu(\xi), \tag{2.2}$$

where $m_\alpha^\nu(\xi) = \psi(2^{-\nu}|\xi|)|\xi|^{-\alpha} e^{\pm i|\xi|}$. Since the function m_α^ν is radial, it follows that $K_\alpha^\nu = \mathcal{F}_k^{-1}(m_\alpha^\nu)$ is radial and from (1.1),

$$K_\alpha^\nu(x) = d_k \int_0^\infty \psi(2^{-\nu}s) s^{-\alpha+2\gamma_k+n-1} e^{is} \mathcal{J}_{\gamma_k+n/2-1}(|x|s)ds.$$

In the next we claim that the sum $\sum K_\alpha^\nu(x)$ is convergent for $|x| \neq 0, 1$ and we have

$$K_\alpha(x) = \sum_{\nu=0}^{\infty} K_\alpha^\nu(x) = d_k \int_0^\infty \widetilde{\phi}(s) s^{-\alpha+2\gamma_k+n-1} e^{is} \mathcal{J}_{\gamma_k+n/2-1}(|x|s) ds$$

For this purpose we recall the appropriate asymptotic expansion of Bessel function (see [16]),

$$\begin{aligned} \mathcal{J}_{\gamma_k+n/2-1}(|x|s) &= e^{i|x|s} \sum_{\ell=0}^{N-1} a_\ell (|x|s)^{-\gamma_k-(n-1)/2-\ell} \\ &\quad + e^{-i|x|s} \sum_{\ell=0}^{N-1} a'_\ell (|x|s)^{-\gamma_k-(n-1)/2-\ell} + R_N(|x|s), \end{aligned} \tag{2.3}$$

where a_ℓ and a'_ℓ are constants and the function R_N satisfies the estimate

$$|R_N(t)| \leq c_N |t|^{-N-\gamma_k-(n-1)/2}. \tag{2.4}$$

Thus for a large enough N, one can write

$$K_\alpha^\nu(x) = d_k \sum_{\ell=0}^{N-1} |x|^{-\gamma_k-(n-1)/2-\ell} \Big\{ a_\ell \, f_\alpha^{\nu,\ell}(1+|x|) + a'_\ell \, f_\alpha^{\nu,\ell}(1-|x|) \Big\} + d_k G_\alpha^{N,\nu}(x)$$

where we put

$$\begin{aligned} f_\alpha^{\nu,\ell}(t) &= \int_0^\infty \psi(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2-\ell} e^{its} ds, \\ G_\alpha^{N,\nu}(x) &= \int_0^\infty \psi(2^{-\nu}s) s^{-\alpha+2\gamma_k+n-1} R_N(|x|s) \, e^{is} \, ds. \end{aligned}$$

After possibly interchanging sum and integral, we have

$$\begin{aligned} \sum_{\nu=0}^{\infty} G_{N,\nu}(x) &= \int_0^\infty \widetilde{\phi}(s) s^{-\alpha+2\gamma_k+n-1} R_N(|x|s) \, e^{is} \, ds, \\ \sum_{\nu=0}^{\infty} f_{\nu,\ell}(t) &= \int_0^\infty \widetilde{\phi}(s) s^{-\alpha+\gamma_k+(n-1)/2-\ell} e^{its} ds; \quad \ell \geq 1. \end{aligned}$$

When for $\ell = 0$ we shall need to integrate by parts,

$$\begin{aligned} &\int_0^\infty \psi(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2} e^{its} ds = it^{-1} \int_0^\infty 2^{-\nu} \psi'(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2} e^{its} ds \\ &+ i(-\alpha+\gamma_k+(n-1)/2) t^{-1} \int_0^\infty \psi(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2-1} e^{its} ds. \end{aligned}$$

The interchange of summation being permissible because

$$\begin{aligned} \sum_{\nu=0}^{\infty} \int_0^\infty |2^{-\nu} \psi'(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2} e^{its}| &= \sum_{\nu=0}^{\infty} \int_0^\infty |2^{-\nu} s \, \psi'(2^{-\nu}s)| s^{-\alpha+\gamma_k+(n-1)/2-1} ds \\ &\leq C \sum_{\nu=0}^{\infty} \int_{2^{\nu-1}}^{2^{\nu+1}} s^{-\alpha+\gamma_k+(n-1)/2-1} ds \\ &\leq 2C \int_{1/2}^\infty s^{-\alpha+\gamma_k+(n-1)/2-1} ds < \infty. \end{aligned}$$

Thus,

$$\sum_{\nu=0}^{\infty}\int_0^{\infty} 2^{-\nu}\psi'(2^{-\nu}s)s^{-\alpha+\gamma_k+(n-1)/2}e^{its}ds=\int_0^{\infty}\widetilde{\phi}'(s)s^{-\alpha+\gamma_k+(n-1)/2}e^{its}ds.$$

and

$$\begin{aligned}\sum_{\nu=0}^{\infty} f_{\alpha}^{\nu,0}(t) &= it^{-1}\int_0^{\infty}\widetilde{\phi}'(s)s^{-\alpha+\gamma_k+(n-1)/2}e^{its}ds\\ &+ i(-\alpha+\gamma_k+(n-1)/2)t^{-1}\int_0^{\infty}\widetilde{\phi}(s)s^{-\alpha+\gamma_k+(n-1)/2-1}e^{its}ds\\ &= \int_0^{\infty}\widetilde{\phi}(s)s^{-\alpha+\gamma_k+(n-1)/2}e^{its}ds.\end{aligned}$$

We therefore conclude that

$$\begin{aligned}K_{\alpha}(x)=d_k\sum_{\ell=0}^{N-1}|x|^{-\gamma_k-(n-1)/2-\ell}\Big\{a_{\ell}\int_0^{\infty}\widetilde{\phi}(s)s^{-\alpha+\gamma_k+(n-1)/2-\ell}e^{is}e^{i|x|s}ds\\ +a'_{\ell}\int_0^{\infty}\widetilde{\phi}(s)s^{-\alpha+\gamma_k+(n-1)/2-\ell}e^{is}e^{-i|x|s}ds.\Big\}+d_k\int_0^{\infty}\widetilde{\phi}(s)s^{-\alpha+2\gamma_k+n-1}R_N(|x|s)\,e^{is}\,ds.\end{aligned}\tag{2.5}$$

and in view of (2.3)

$$K_{\alpha}(x)=d_k\int_0^{\infty}\widetilde{\phi}(s)s^{-\alpha+2\gamma_k+n-1}e^{is}\mathcal{J}_{\gamma_k+n/2-1}(|x|s)ds.$$

In order to study the behavior of K_{α} we will need the following elementary lemmas.

Lemma 2.1. *Let $\alpha>0$. The function given by*

$$h(t)=\int_0^{\infty}\phi(s)s^{-\alpha}e^{\pm its}ds,\qquad t\neq 0$$

satisfies $h(t)=o(t^{-N})$ when $t\to\infty$, for any integer $N>0$.

Proof. By using integration by parts and Leibniz rule,

$$\begin{aligned}|t^N h(t)| &= \left|\int_0^{\infty}\left(\frac{d}{ds}\right)^N\Big(\phi(s)s^{-\alpha}\Big)e^{\pm its}ds\right|\\ &\leq C\left\{\int_{1/2}^1\sum_{\ell=0}^{N}|\,\phi^{(\ell)}(s)s^{-\alpha-N+\ell}|ds+\int_1^{\infty}s^{-\alpha-N}ds\right\}\\ &\leq C\,.\end{aligned}$$

□

Lemma 2.2. *The following are continuous and bounded functions on $(0,\infty)$*

$$G(t)=\int_0^t s^{-\alpha+\gamma_k+(n-1)/2}e^{\pm is}\,ds,\;\; H(t)=t^{-\alpha+\gamma_k+(n+1)/2}\int_1^2\psi(s)s^{-\alpha+\gamma_k+(n-1)/2}e^{\pm its}\,ds.$$

Proof. The continuous of G and H is obvious. Since the integral defining G is convergent then the function G has a finite limit at ∞, from which and continuity we obtain the boundedness of G. Similarly for the boundedness of H, since by integration by parts we have that

$$\left|\int_1^2 \psi(s)s^{-\alpha+\gamma_k+(n-1)/2}\, e^{\pm its}\, ds\right| \leq Ct^{-1},$$

and then $\lim_{t\to\infty} H(t) = 0$. □

Let us now consider the asymptotic behavior of K_α.

Lemma 2.3. *The kernel K_α has the following properties:*

(i) $K_\alpha(x) = O(|x|^{-N}), \qquad as\ |x| \to \infty,\ for\ big\ N,$

(ii) $K_\alpha(x) = O\Big((1-|x|)^{\alpha-\gamma_k-(n-1)/2-1}\Big), \qquad as\ |x| \to 1,$

(iii) $K_\alpha(x)$ *is bounded near the origin,*

(iv) K_α *is an integrable function.*

Proof. Clearly (iv) is a consequence of (i), (ii) and (iii). The proof of (i) follows from (2.5), Lemma 2.1 and (2.4). Consider (2.5) we see that except the term corresponding to $\ell = 0$ all terms are bounded near $|x| = 1$. On the other hand, writing

$$\begin{aligned}
\int_0^\infty \widetilde{\phi}(s)s^{-\alpha+\gamma_k+(n-1)/2}e^{i(1\pm|x|)s}ds &= \int_0^\infty s^{-\alpha+\gamma_k+(n-1)/2}e^{i(1\pm|x|)s}ds \\
&\quad + \int_0^\infty (\widetilde{\phi}(s)-1)s^{-\alpha+\gamma_k+(n-1)/2}e^{i(1\pm|x|)s}ds \\
&= (1\pm|x|)^{\alpha-\gamma_k-(n-1)/2-1}\int_0^\infty s^{-\alpha+\gamma_k+(n-1)/2}e^{is}ds \\
&\quad + \int_0^\infty (\widetilde{\phi}(s)-1)s^{-\alpha+\gamma_k+(n-1)/2}e^{i(1\pm|x|)s}ds \\
&= C(1\pm|x|)^{\alpha-\gamma_k-(n-1)/2-1} + H(x),
\end{aligned}$$

where H is bounded. This gives the property (ii). To prove (iii) we can use the well known integral representation for Bessel function to write

$$\begin{aligned}
K_\alpha^\nu(x) &= d_k\int_0^\infty \psi(2^{-\nu}s)s^{-\alpha+2\gamma_k+n-1}e^{is}\mathcal{J}_{\gamma_k+n/2-1}(|x|s)ds \\
&= b_k\int_{-1}^1 (1-t^2)^{\gamma_k+(n-3)/2}\left\{\int_0^\infty \psi(2^{-\nu}s)s^{-\alpha+2\gamma_k+n-1}e^{i(1+t|x|)s}\, ds\right\}\, dt,
\end{aligned} \tag{2.6}$$

where $b_k = d_k\Gamma(\gamma_k+n/2)/\Gamma(\gamma_k+(n+1)/2-1)$. Let $|x| \leq 1/2$ and $N > -\alpha+2\gamma_k+n$, sufficiently large. Using the fact that

$$\left|\left(\frac{d}{ds}\right)^\ell (\psi(2^{-\nu}s))\right| \leq C|s|^{-\ell}, \quad \ell \in \mathbb{N}$$

and applying integration by parts N times, to derive that

$$\begin{aligned}
&\int_0^\infty \psi(2^{-\nu}s)s^{-\alpha+2\gamma_k+n-1}e^{i(1+t|x|)s}\\
&= (i(1+t|x|))^{-N}\int_0^\infty \left(\frac{d}{ds}\right)^N \Big(\psi(2^{-\nu}s)s^{-\alpha+2\gamma_k+n-1}\Big)e^{i(1+t|x|)s}ds\\
&\leq C2^{-N}\int_{2^{\nu-1}}^{2^{\nu+1}} s^{-\alpha+2\gamma_k+n-1-N}ds.
\end{aligned}$$

From which and (2.6) it follows that

$$|K_\alpha^\nu(x)| \leq C\int_{2^{\nu-1}}^{2^{\nu+1}} s^{-\alpha+2\gamma_k+n-1-N}ds \tag{2.7}$$

and then $|K_\alpha(x)| \leq C$. This concludes (iii). □

Lemma 2.4. *Let $\alpha \in]\gamma_k + (n-1)/2, \gamma_k + (n+1)/2[$. The operator $T_{m\alpha}$ can be represented through the kernel K_α as the integral,*

$$T_{m_\alpha}(f)(x) = \int_{\mathbb{R}^n} K_\alpha(y)\tau_x(f)(y)w_k(y)dy, \qquad f \in S(\mathbb{R}^n) \tag{2.8}$$

and satisfies the inequality

$$\|T_{m_\alpha}(f)\|_{1,k} \leq \|K_\alpha\|_{1,k}\|f\|_{1,k}. \tag{2.9}$$

Proof. Let $f \in S(\mathbb{R}^n)$. In view of (2.2) we can write

$$T_{m_\alpha} = \sum_{\nu=0}^\infty T_{m_\alpha^\nu}\,.$$

Each of $T_{m_\alpha^\nu}$ will be written in its integral form

$$T_{m_\alpha^\nu}(f)(x) = K_\alpha^\nu *_k f(x) = \int_{\mathbb{R}^n} K_\alpha^\nu(y)\tau_x(f)(-y)w_k(y)dy.$$

Hence to establish (2.8) we only need to interchange the order of integration and summation. For this purpose we split the integral

$$\begin{aligned}
\int_{\mathbb{R}^n} K_\alpha^\nu(y)\tau_x(f)(y)w_k(y)dy &= \int_{|y|\leq 1/2} K_\alpha^\nu(y)\tau_x(f)(y)w_k(y)dy\\
&+ \int_{1/2\leq|y|\leq 2} K_\alpha^\nu(y)\tau_x(f)(y)w_k(y)dy + \int_{|y|\geq 2} K_\alpha^\nu(y)\tau_x(f)(y)w_k(y)dy.
\end{aligned}$$

From the estimate (2.7) we have

$$\sum_{\nu=0}^\infty \int_{|y|\leq 1/2} K_\alpha^\nu(y)\tau_x(f)(y)w_k(y)dy = \int_{|y|\leq 1/2} K_\alpha(y)\tau_x(f)(y)w_k(y)dy.$$

Similarly for the integral over $|y| \geq 2$, by writing

$$K_\alpha^\nu(y) = \sum_{\ell=0}^{N-1} |y|^{-\gamma_k-(n-1)/2-\ell} \Bigg\{ c_\ell \int_0^\infty \psi(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2-\ell} e^{i(1+|y|)s} ds$$

$$+ c'_\ell \int_0^\infty \psi(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2-\ell} e^{i(1-|y|)s} ds \Bigg\} + \int_0^\infty \psi(2^{-\nu}s) s^{-\alpha+2\gamma_k+n-1} R_N(|y|s)\, e^{is}\, ds,$$

for N sufficiently large, we get that

$$|K_\alpha^\nu(y)| \leq C \int_{2^{\nu-1}}^{2\nu+1} s^{-\alpha+\gamma_k+(n-1)/2-1}\, ds.$$

Notice that we have integrate by parts the integral of first term (for $\ell = 0$). Thus,

$$\sum_{\nu=0}^{\infty} \int_{|y|\geq 2} K_\alpha^\nu(y) \tau_x(f)(y) w_k(y) dy = \int_{|y|\geq 2} K_\alpha(y) \tau_x(f)(y) w_k(y) dy.$$

Now for the integral over $1/2 \leq |y| \leq 2$ we proceed by applying the dominated convergence theorem. Put

$$\Psi_\nu(s) = \sum_{j=0}^{\nu} \psi(2^{-j}s), \qquad S_\alpha^\nu(y) = \sum_{j=0}^{\nu} K_\alpha^\nu(y).$$

First, observe that $\Psi_\nu(s) = 1$ if $1 \leq s \leq 2^\nu$ and $\Psi_\nu(s) = 0$ if $s \geq 2^{\nu+1}$ or $s \leq 1/2$. Using this fact, we have

$$\int_0^\infty \Psi_\nu(s) s^{-\alpha+\gamma_k+(n-1)/2} e^{i(1\pm|y|)s} ds$$

$$= \int_0^1 \psi(s) s^{-\alpha+\gamma_k+(n-1)/2} e^{i(1\pm|y|)s} ds + \int_1^{2^\nu} s^{-\alpha+\gamma_k+(n-1)/2} e^{i(1\pm|y|)s} ds$$

$$+ \int_{2^\nu}^{2^{\nu+1}} \psi(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2} e^{i(1\pm|y|)s} ds.$$

Clearly

$$\left| \int_0^1 \psi(s) s^{-\alpha+\gamma_k+(n-1)/2} e^{i(1\pm|y|)s} ds \right| \leq \int_0^1 \psi(s) s^{-\alpha+\gamma_k+(n-1)/2} ds. \leq C$$

However, Lemma 2.2 implies that

$$\left| \int_1^{2^\nu} s^{-\alpha+\gamma_k+(n-1)/2} e^{i(1\pm|y|)s} ds \right| = |1 \pm |y||^{\alpha-\gamma_k-(n+1)/2} \left| \int_{|1\pm|y||}^{2^\nu|1\pm|y||} s^{-\alpha+\gamma_k+(n-1)/2} e^{\pm is} ds \right|$$

$$\leq C\, |1 \pm |y||^{\alpha-\gamma_k-(n+1)/2}$$

and

$$\left| \int_{2^\nu}^{2^{\nu+1}} \psi(2^{-\nu}s) s^{-\alpha+\gamma_k+(n-1)/2} e^{i(1\pm|y|)s} ds. \right| = |1 \pm |y||^{\alpha-\gamma_k-(n+1)/2} H\Big((1 \pm |y|)2^\nu\Big)$$

$$\leq C\, |1 \pm |y||^{\alpha-\gamma_k-(n+1)/2}.$$

Now writing

$$S_\nu(y)=\sum_{p=0}^{N-1}|y|^{-\gamma_k-(n-1)/2-p}\Big\{c_p\int_0^\infty\Psi_\nu(s)s^{-\alpha+\gamma_k+(n-1)/2-p}e^{i(1+|y|)s}ds$$
$$+c'_p\int_0^\infty\Psi_\nu(s)s^{-\alpha+\gamma_k+(n-1)/2-p}e^{i(1-|y|)s}ds\Big\}+\int_0^\infty\Psi_\nu(s)s^{-\alpha+2\gamma_k+n-1}R_N(|y|s)\,e^{is}\,ds.$$

So, as $1/2\le|y|\le 2$ we get that

$$|S_\nu(y)|\le C\,|1-|y||^{\alpha-\gamma_k-(n+1)/2}. \tag{2.10}$$

Since the right hand side in (2.10) is an integrable function then by the dominated convergence theorem,

$$\sum_{\nu=0}^{\infty}\int_{1/2\le|y|\le 2}K_\alpha^\nu(y)\tau_x(f)(y)w_k(y)dy=\int_{1/2\le|y|\le 2}K_\alpha(y)\tau_x(f)(y)w_k(y)dy.$$

This completes the proof of (2.8).

To prove (2.9), we consider the truncated kernels and operators, for $0<\varepsilon<1$

$$K_{\alpha,\varepsilon}=K_\alpha\,\mathbb{1}_{\{\varepsilon\le||y|-1|\le\frac{1}{\varepsilon}\}}$$

and for $f\in S(\mathbb{R}^n)$,

$$T_{m_\alpha,\varepsilon}f(x)=\int_{\mathbb{R}^n}K_{\alpha,\varepsilon}(y)\tau_x(f)(y)w_k(y)dy=\int_{\mathbb{R}^n}\tau_x(K_{\alpha,\varepsilon})(y)f(y)w_k(y)dy.$$

Since the kernel K_α is in $L^1(\mathbb{R}^n,w_k(y)dy)$ and bounded a way from $|y|=1$ which imply that $K_{\alpha,\varepsilon}$ is a bounded radial function in $L^1(\mathbb{R}^n,w_k(y)dy)$, we can then use (1.4) and (1.2) to get the following

$$\|T_{m_\alpha,\varepsilon}(f)\|_{1,k}\le\|\tau_x(K_{\alpha,\varepsilon})\|_{1,k}\|f\|_{1,k}\le\|K_{\alpha,\varepsilon}\|_{1,k}\|f\|_{1,k}\le\|K_\alpha\|_{1,k}\|f\|_{1,k}.$$

As $\lim_{\varepsilon\to\infty}T_{m_\alpha,\varepsilon}(f)(x)=T_{m_\alpha}(f)(x)$, then we can apply Fatou's lemma to obtain (2.9). □

Now, we are in a position to prove Theorem 1.1.

Proof of Theorem 1.1 . We can assume that $1\le p\le 2$, the case $2\le p\le\infty$ follows by a well known duality argument. Let's choose an arbitrary $\alpha_0\in\,]\gamma_k+(n-1)/2,\gamma_k+(n+1)/2[$. We first write

$$\begin{aligned}m_\alpha^\nu(\xi)&=\Big\{\psi(2^{-\nu}|\xi|)|2^{-\nu}\xi|^{\alpha_0-\alpha}\Big\}\Big\{e^{i|\xi|}|\xi|^{-\alpha_0}\phi(|\xi|)\Big\}\\&=2^{\nu(\alpha_0-\alpha)}h(2^{-\nu}\xi)m_{\alpha_0}(\xi)\end{aligned}$$

where $h(\xi)=\psi(|\xi|)|\xi|^{\alpha_0-\alpha}$. The fact that h is a radial C^∞-function with compact support, the corresponding multiplier operator T_h is the convolution operator with the radial Schwartz function $\mathcal{F}_k^{-1}(h)$ and therefore bounded on $L^r(\mathbb{R}^n,w_k(x)dx)$ for all $1\le$

$r \leq \infty$. This follows from (1.4). Also, a simple argument shows that the multiplier T_{h_ν} of symbol $h_\nu(\xi) = h(2^{-\nu}\xi)$ is bounded on $L^r(\mathbb{R}^n, w_k(x)dx)$ with norm $\|T_{h_\nu}\|_r = \|T_h\|_r$. Now, using Lemma 2.9 and that $T_{m_\alpha^\nu}$ is a composition of two multiplier operators on $L^1(\mathbb{R}^n, w_k(x)dx)$, it follows that $T_{m_\alpha^\nu}$ is bounded on $L^1(\mathbb{R}^n, w_k(x)dx)$ with norm

$$\|T_{m_\alpha^\nu}\|_1 \leq C2^{\nu(\alpha_0-\alpha)}.$$

Similarly if we write $m_\alpha^\nu(\xi) = 2^{-\alpha\nu}\Big\{\psi(2^{-\nu}|\xi|)|2^{-\nu}\xi|^{-\alpha}\Big\}\Big\{e^{i|\xi|}\phi(|\xi|)\Big\}$ then we get that

$$\|T_{m_\nu}\|_2 \leq C2^{-\alpha\nu}.$$

Therefore, by the Riesz-Thorin interpolation theorem, $T_{m_\alpha^\nu}$ is bounded on $L^p(\mathbb{R}^n, w_k(x)dx)$ for any $1 \leq p \leq 2$, $1/p = 1 - \theta/2$, $\theta \in [0,1]$, and we have that

$$\|T_{m_\alpha^\nu}\|_p \leq C2^{\nu(\alpha_0-\alpha)(1-\theta)-\alpha\nu\theta} = C2^{\nu[\alpha_0(1-\theta)-\alpha]}. \tag{2.11}$$

We can then sum and obtain the boundedness of T_{m_α} on $L^p(\mathbb{R}^n, w_k(x)dx)$ if $\alpha > \alpha_0(1-\theta)$. Since, α_0 is arbitrary in $]\gamma_k + (n-1)/2,\ \gamma_k + (n+1)/2[$, it suffices to take

$$\alpha > \gamma_k + (n-1)/2(1-\theta) = (2\gamma_k + (n-1))\left(\frac{1}{p} - \frac{1}{2}\right).$$

Our theorem is therefore proved. □

2.2 Proof of Theorem 1.4

We will use the following technical Lemma

Lemma 2.5. *Suppose that F is $C^1(I)$, for an interval I. Then for each $0 < \lambda \leq |I|$ and $p, p' > 1$ with $1/p + 1/p' = 1$, we have*

$$\sup_I |F(t)| \leq \lambda^{-1/p}\left(\int_I |F(t)|^p dt\right)^{1/p} + \lambda^{1/p'}\left(\int_I |F'(t)|^p dt\right)^{1/p} \tag{2.12}$$

Proof. Let $I_0 \subset I$ be an interval of length λ. For $t, s \in I_0$ we have

$$F(t) = F(s) + \int_s^t F'(u)du.$$

So by Hölder's inequality

$$|F(t)| \leq |F(s)| + \lambda^{1/p'}\left(\int_I |F'(t)|^p dt\right)^{1/p}.$$

Integrating both sides with respect to s, we get

$$|F(t)| \leq \lambda^{-1}\int_I |F(s)|ds + \lambda^{1/p'}\left(\int_I |F'(t)|^p dt\right)^{1/p}$$

and using again Holder's inequality to obtain

$$|F(t)| \leq \lambda^{-1/p}\left(\int_I |F(t)|^p dt\right)^{1/p} + \lambda^{1/p'}\left(\int_I |F'(t)|^p dt\right)^{1/p}.$$

Then (2.12) is established. □

The next lemma is essentially contained in the theorems 6.2 and 6.1 of [14].

Lemma 2.6. *Let $\Phi \in L^1(\mathbb{R}^n w_k(x)dx)$ be a real valued radial function with $\mathcal{F}_k(\Phi) \in L^1(\mathbb{R}^n, w_k(x)dx)$ and which satisfies $|\Phi(x)| \leq C(1+|x|)^{-2\gamma_k-n-1}$. Then we have*

$$\| \sup_{t>0} \Phi_t *_k f(x)\|_{p,k} \leq C\|f\|_{p,k}; \quad 1 < p \leq \infty,$$

where $\Phi_t(x) = t^{-2\gamma_k-n}\Phi(t^{-1}x)$.

Now we consider our maximal operator

$$A_\alpha(f)(x) = \sup_{t>0} |A_\alpha(f)(x,t)|$$

where

$$A_\alpha(f)(x,t) = \int_{\mathbb{R}^n} \frac{e^{\pm it|\xi|}}{|t\xi|^\alpha} \phi(t\xi)\mathcal{F}(f)(\xi)E_k(ix,\xi)w_k(\xi)d\xi.$$

We write

$$A_\alpha(f)(x,t) = \sum_{\nu=0}^{\infty} A_\alpha^\nu(f)(x,t)$$

where

$$A_\alpha^\nu(f)(x,t) = \int_{\mathbb{R}^n} \frac{e^{\pm it|\xi|}}{|t\xi|^\alpha} \phi(t\xi)\psi(2^{-\nu}t|\xi|)\mathcal{F}(f)(\xi)E_k(ix,\xi)w_k(\xi)d\xi.$$

Let $t \in [1,2]$. We can use the same argument that provided (2.11) and duality to get the following estimates

$$\|A_\alpha^\nu(f)(.,t))\|_{p,k} \leq C2^{\nu[\alpha_0|2/p-1|-\alpha]}\|f\|_{p,k}$$

$$\|\frac{\partial}{\partial t}A_\alpha^\nu(f)(.,t)\|_{p,k} \leq C2^{\nu[\alpha_0|2/p-1|-(\alpha-1)]}\|f\|_{p,k}$$

for an arbitrary $\alpha_0 \in]\gamma_k + (n-1)/2,\ \gamma_k + (n+1)/2$ and $p \geq 1$. Now by applying the lemma 2.5 with $\ell = 2^{-\nu}$ it follows that

$$\| \sup_{1\leq t\leq 2} A_\alpha^\nu(f)(.,t)\|_{p,k} \leq C2^{\nu[\alpha_0|2/p-1|-\alpha+1/p]}\|f\|_{p,k}\ . \tag{2.13}$$

In addition, (2.13) implies that for $j \in \mathbb{Z}$,

$$\| \sup_{2^j\leq t\leq 2^{j+1}} A_\alpha^\nu(f)(.,t)\|_{p,k} \leq C2^{\nu[\alpha_0|2/p-1|-\alpha+1/p]}\|f\|_{p,k}$$

which can be seen by writing

$$\sup_{2^j\leq t\leq 2^{j+1}} |A_\alpha^\nu(f)(x,t)| = \sup_{1\leq t\leq 2} |A_\alpha^\nu(f(2^j.))(2^{-j}x,t)|. \tag{2.14}$$

In the next we claim that for $p \geq 2$,

$$\| \sup_{t>0} |A_\alpha^\nu(f)(.,t)\|_{p,k} \leq C\ 2^{\nu[\alpha_0|2/p-1|-\alpha+1/p]}\|f\|_{p,k} \tag{2.15}$$

which asserts that

$$\| \sup_{t>0} |A_\alpha(f)(.,t)\|_{p,k} \leq C \, \|f\|_{p,k}, \tag{2.16}$$

for $p \geq 2$ and $\alpha > (2\gamma_k + (n-1))\,(1/2 - 1/p) + 1/p$. For the proof we invoke a variant of the Littlewood-Paley operators associated to Dunkl transform , see Corollary 4.2 of [2]. Let $t \in [2^j, 2^{j+1}]$, $j \in \mathbb{Z}$. Define the function f_ℓ by $\mathcal{F}_k(f_\ell)(\xi) = \mathcal{F}_k(f)(\xi)\psi(2^{-\ell}|\xi|)$, $\ell \in \mathbb{Z}$. It follows that

$$A^\nu_\alpha(f)(x,t) = A^\nu_\alpha \left(\sum_{|\ell+j-\nu|\leq 2} f_\ell \right)(x,t).$$

From this and (2.14) we have for all $p \geq 2$

$$\begin{aligned}
\int_{\mathbb{R}^n} \sup_{t>0} |A^\nu_\alpha(f)(x,t)|^p w_k(x)dx &\leq \sum_{j\in\mathbb{Z}} \int_{\mathbb{R}^n} \sup_{2^j\leq t\leq 2^{j+1}} |A^\nu_\alpha(f)(x,t)|^p w_k(x)dx \\
&\leq C\, 2^{\nu p\,[\alpha_0|2/p-1|-\alpha+1/p]} \sum_{j\in\mathbb{Z}} \int_{\mathbb{R}^n} \sum_{|\ell+j-\nu|\leq 2} |f_\ell(x)|^p w_k(x)dx \\
&\leq C\, 2^{\nu p\,[\alpha_0|2/p-1|-\alpha+1/p]} \int_{\mathbb{R}^n} \sum_{\ell\in\mathbb{Z}} |f_\ell(x)|^p w_k(x)dx \\
&\leq C\, 2^{\nu p[\alpha_0|2/p-1|-\alpha+1/p]} \int_{\mathbb{R}^n} \left(\sum_{\ell\in\mathbb{Z}} |f_\ell(x)|^2 \right)^{p/2} w(x)dx \\
&\leq C\, 2^{\nu p[\alpha_0|2/p-1|-\alpha+1/p]} \|f\|^p_{p,k}.
\end{aligned}$$

This yields the claim (2.15).

The range $p \leq 2$ follows by interpolation argument, we proceed as in the proof of the theorem 1.2. Let us first observe that when $\alpha > \gamma_k + (n+1)/2$ the kernel K_α of the operator T_{m_α} satisfies the following decay estimates

$$|K_\alpha(x)| \leq \frac{C_N}{(1+|x|)^N}; \quad n \in \mathbb{N}, \qquad N \in \mathbb{N}$$

which is immediately by making use of (1.1), (2.3) and integration by parts. Hence using Lemma 2.6 we get the boundedness of A_α on $L^p(\mathbb{R}^n, w(x)dx)$ for $1 < p \leq \infty$. The key tool is the following. For an arbitrary $\alpha_0 > \gamma_k + (n+1)/2$, one can write

$$\begin{aligned}
m^\nu_\alpha(t\xi) &= \Big\{ \psi(2^{-\nu}t|\xi|)|2^{-\nu}t\xi|^{\alpha_0-\alpha} \Big\} \Big\{ e^{it|\xi|}|t\xi|^{-\alpha_0}\phi(t\xi) \Big\} \\
&= 2^{\nu(\alpha_0-\alpha)} h(2^{-\nu}t\xi) m_{\alpha_0}(t\xi)
\end{aligned}$$

where $h(\xi) = \psi(|\xi|)|\xi|^{\alpha_0-\alpha}$. Put $H = \mathcal{F}_k^{-1}(h)$, which is a radial Schwartz function. Thus we can write

$$A^\nu_\alpha(f)(x,t) = 2^{\nu(\alpha_0-\alpha)} H_{2^\nu t^{-1}} *_k A_{\alpha_0}(f)(.,t)(x)$$

and we have that

$$|A^\nu_\alpha(f)(x,t)| \leq 2^{\nu(\alpha_0-\alpha)} |H_{2^\nu t^{-1}}| *_k |A_{\alpha_0}(f(.,t)|(x) \leq 2^{\nu(\alpha_0-\alpha)} |H|_{2^\nu t^{-1}} *_k |A_{\alpha_0}(f)|(x).$$

Then from Lemma 2.6 and the boundedness of A_{α_0} we get

$$\|\sup_{t>0}|A^{\nu}_{\alpha}(f)(.,t)\|_{q,k}\leq C2^{\nu(\alpha_0-\alpha)}\|f\|_{q,k};\quad 1<q\leq\infty.$$

On the other hand, the boundedness on $L^2(\mathbb{R},w_k(x)dx)$ in (2.15) gives

$$\|\sup_{t>0}|A^{\nu}_{\alpha}(f)(.,t)\|_{2,k}\leq C2^{\nu(1/2-\alpha)}\|f\|_{2,k}.$$

So, by using the Riesz-Thorin interpolation Theorem,

$$\|\sup_{t>0}|A^{\nu}_{\alpha}(f)(.,t)\|_{p}\leq C\ 2^{\nu(\theta(\alpha_0-\alpha)+(1-\theta)(1/2-\alpha))}=C\ 2^{\nu(\theta(\alpha_0-1/2)+1/2-\alpha)}\qquad(2.17)$$

for all $1<q\leq p\leq 2$, where $1/p=1/2+\theta(1/q-1/2)$. It is now easy to check from this that the sum $\sum_{\nu}\|\sup_{t>0}|A^{\nu}_{\alpha}(f)(.,t)\|_p$ is finite when

$$\alpha>(\frac{1}{p}-\frac{1}{2})(2\gamma_k+n-1)+\frac{1}{p}.$$

In fact, as $\theta=(1/p-1/2)(1/q-1/2)^{-1}$ we see that $\theta(\alpha_0-1/2)+1/2$ tends to $(1/p-1/2)(2\gamma_k+n-1)+1/p$ when letting α_0 go to $\gamma_k+(n+1)2$ and q go to 1 which guarantees the choice of α_0 and q such that $\alpha>\theta(\alpha_0-1/2)+1/2$. This conclude the case $p\leq 2$ and thus we have completely proved our theorem.

2.3 Proof of Theorem 1.5

Let $\alpha>0$. As a first step we write

$$\begin{aligned}\mathcal{J}_{\alpha+\gamma_k+n/2-1}(t|\xi|) &= \mathcal{J}_{\alpha+\gamma_k+n/2-1}(t|\xi|)\phi(t\xi)+\mathcal{J}_{\alpha+\gamma_k+n/2-1}(t|\xi|)(1-\phi(t\xi))\\ &= a^{(1)}_{\alpha}(t\xi)+a^{(2)}_{\alpha}(t\xi)\end{aligned}$$

and so, $\mathcal{M}_{\alpha}(f)=\mathcal{M}^{(1)}_{\alpha}+\mathcal{M}^{(1)}_{\alpha}$ with

$$\mathcal{M}^{(i)}_{\alpha}(f)=\sup_{t>0}\left|\int_{\mathbb{R}^n}a^{(i)}_{\alpha}(t|\xi|)\mathcal{F}_k(f)(\xi)w_k(\xi)d\xi\right|=\sup_{t>0}|\mathcal{M}^{(i)}_{\alpha}(f)(x,t)|,\quad i=1,2.$$

Since the function $a^{(2)}_{\alpha}$ is a C^{∞} with compact support, in view of Lemma 2.6 we get the L^p-boundedness of $\mathcal{M}^{(2)}_{\alpha}$ for all $1<p\leq\infty$. To treat the boundedness of $\mathcal{M}^{(1)}_{\alpha}$ we invoke the asymptotic form (2.3) for the Bessel function $\mathcal{J}_{\alpha+\gamma_k+n/2-1}$. For N big enough,

$$\begin{aligned}\mathcal{J}_{\alpha+\gamma_k+n/2-1}(t|\xi|) &= e^{it|\xi|}\sum_{\ell=0}^{N-1}c_{\ell}(t|\xi|)^{-\alpha-\gamma_k-(n-1)/2-\ell}\\ &\quad+e^{-it|\xi|}\sum_{\ell=0}^{N-1}c'_{\ell}(t|\xi|)^{-\alpha-\gamma_k-(n-1)/2-\ell}+R_N(t|\xi|),\end{aligned}$$

where c_{ℓ} are constants. The function R_N satisfies for all $j\in\mathbb{N}$

$$|\partial^j_s R_N(s)|\leq C|s|^{-\alpha-N-\gamma_k-(n-1)/2}.\qquad(2.18)$$

It is not hard to show that this requirement is satisfied. Set $V = \mathcal{F}_k^{-1}(R_N(|.|)\phi(.))$, which can be represented by

$$V(x) = \frac{2^{-\gamma_k - n/2+1}}{\Gamma(\gamma_k + n/2)} \int_0^\infty R_N(s)\widetilde{\phi}(s) s^{2\gamma_k+n-1} \mathcal{J}_{\gamma_k+n/2-1}(|x|s)ds.$$

So, with the use of asymptotic form (2.3), integration by part and (2.18) we get that for some larger N,

$$|V(x)| \leq C(1+|x|)^{-N}$$

and because of Lemma 2.6

$$\| \sup_{t>0} V_t *_k f\|_{p,k} \leq C_p \|f\|_{p,k}; \quad 1 < p \leq \infty.$$

It therefore only remains to check the needed conditions for the boundedness of the multipliers $A_{\alpha+\gamma_k+(n-1)/2+\ell}$, for $\ell = 0, 1...N-1$. But this follow from Theorem 1.4 and can be reduced to the L^p- boundedness of $A_{\alpha+\gamma_k+(n-1)/2}$ that is

$$\alpha + \gamma_k + (n-1)/2 > (2\gamma_k + n - 1)\left|\frac{1}{p} - \frac{1}{2}\right| + \frac{1}{p}$$

or equivalently

(a) $\alpha > 1 - 2\gamma_k - n + (2\gamma_k + n)/p$, if $1 < p \leq 2$

(b) $\alpha > (2 - 2\gamma_k - n)/p$, if $2 \leq p < \infty$,

which is the desired statement of Theorem 1.5.

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